Indian Sign Language Recognition using Machine Learning

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Abstract

This research presents a sign language detection system integrated into a React Native teaching app. Utilizing the “INCLUDE” dataset from Zenodo, it processed video files to extract frames and landmarks for sign gestures. Data preparation involved normalization techniques like Standard Scaler, Label Encoder, and SMOTE. Various models, including LSTM, GRU, and CNN+LSTM, were tested, with the hybrid CNN+LSTM achieving 93.89% accuracy and 94% precision, recall, and F1 scores. The model was deployed via a Flask server, enabling real-time sign prediction within the app.

Introduction

Sign language is a crucial communication tool for the deaf and hard of hearing, yet a communication gap persists, particularly in linguistically diverse countries like India, where Indian Sign Language (ISL) is common. Vision-based Sign Language Recognition (SLR) systems [1], using tools like Mediapipe, OpenCV, [2] and deep learning models such as CNNs and LSTMs, have shown promise in bridging this gap. This study presents a motion-based ISL recognition system [3] leveraging LSTM models and image processing to improve gesture prediction [4] and promote inclusivity for India’s hearing-impaired community.

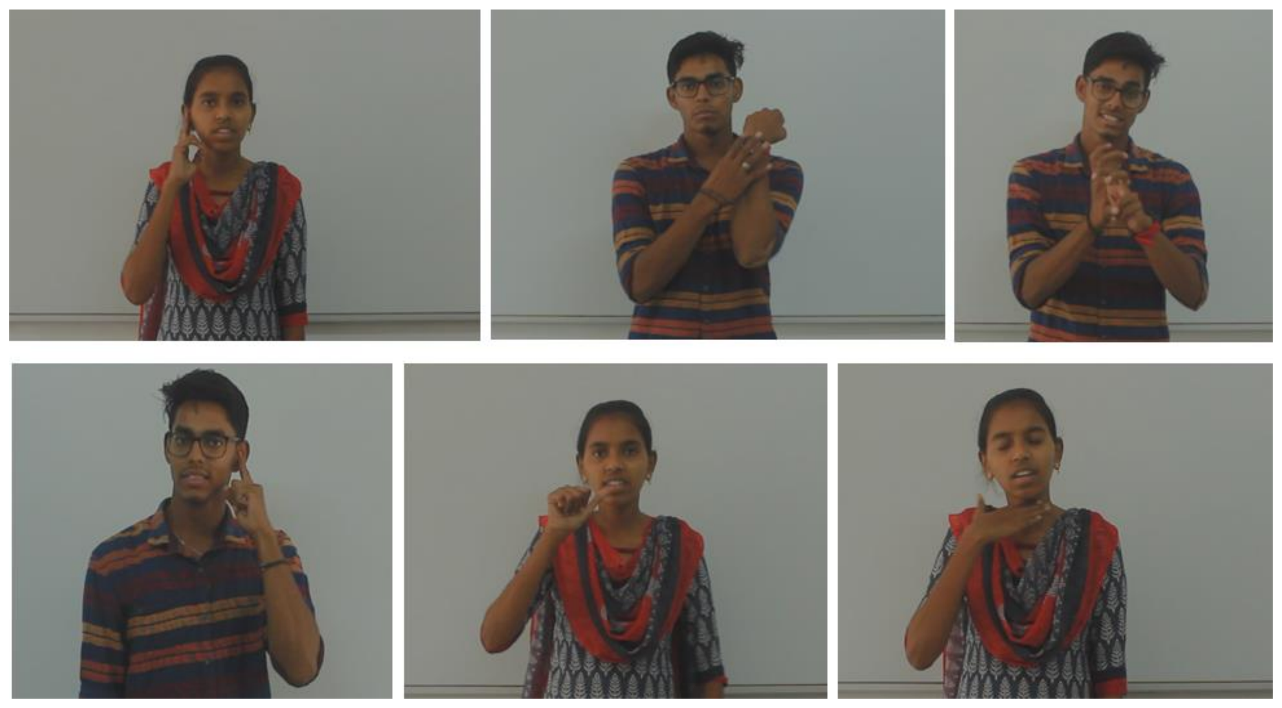

Figure 1. *Sample images from the “Include” dataset*

Methodology

The “INCLUDE” dataset, consisting of MOV video files with Indian Sign Language (ISL) gestures, was used. Video frames were extracted, and landmarks were identified using MediaPipe, and OpenCV. Data preprocessing included normalization with Standard Scaler, Label Encoder, and handling class imbalance with SMOTE. Models tested included LSTM, GRU, and a CNN+LSTM hybrid. The best-performing CNN+LSTM model was integrated into a React Native app, hosted by a Flask server, allowing real-time sign language prediction for users to practice ISL gestures.

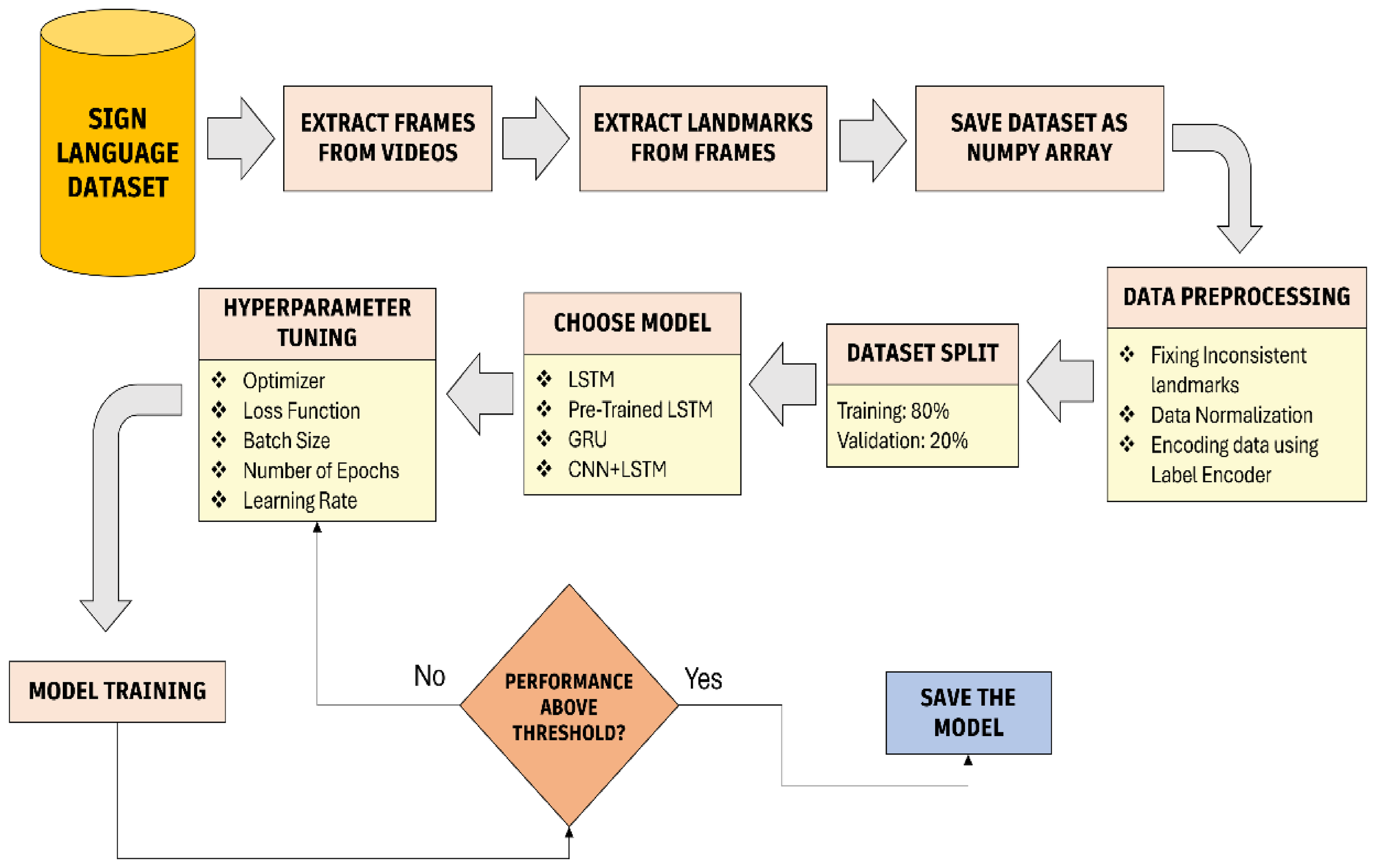

Figure 2. *Flowchart of the methodology*

Results and Discussions

The models were evaluated for accuracy over different epochs. The initial LSTM models which were built using PyTorch and TensorFlow achieved validation accuracies of 79.73% and 85.82%, respectively, with TensorFlow showing better training accuracy and performance metrics. The pre-trained MobileNetV2 reached a validation accuracy of 90.16%, and the GRU model achieved 91.66%, but its precision and recall were lower due to handling long-sequence data poorly. The CNN+LSTM model outperformed all others, achieving 93.61% validation accuracy in just 64 epochs, with high training accuracy and excellent precision, recall, and F1 scores. It also demonstrated the best stability and performance with real-time video inputs.

Models	No. of epochs	Training Accuracy	Training Loss	Validation Accuracy	Validation Loss
LSTM Model 1	594	0.8322	0.4166	0.8165	0.4653
LSTM Model 2	963	0.9077	0.298	0.8966	0.4078
Pre-Trained LSTM (MobileNetV2)	1196	0.9617	0.1243	0.9191	0.3579
GRU	521	0.9558	0.1321	0.9208	0.2841
CNN+LSTM	59	0.9782	0.0772	0.9389	0.1829

Table 1. *Applied models and their performance metrics*

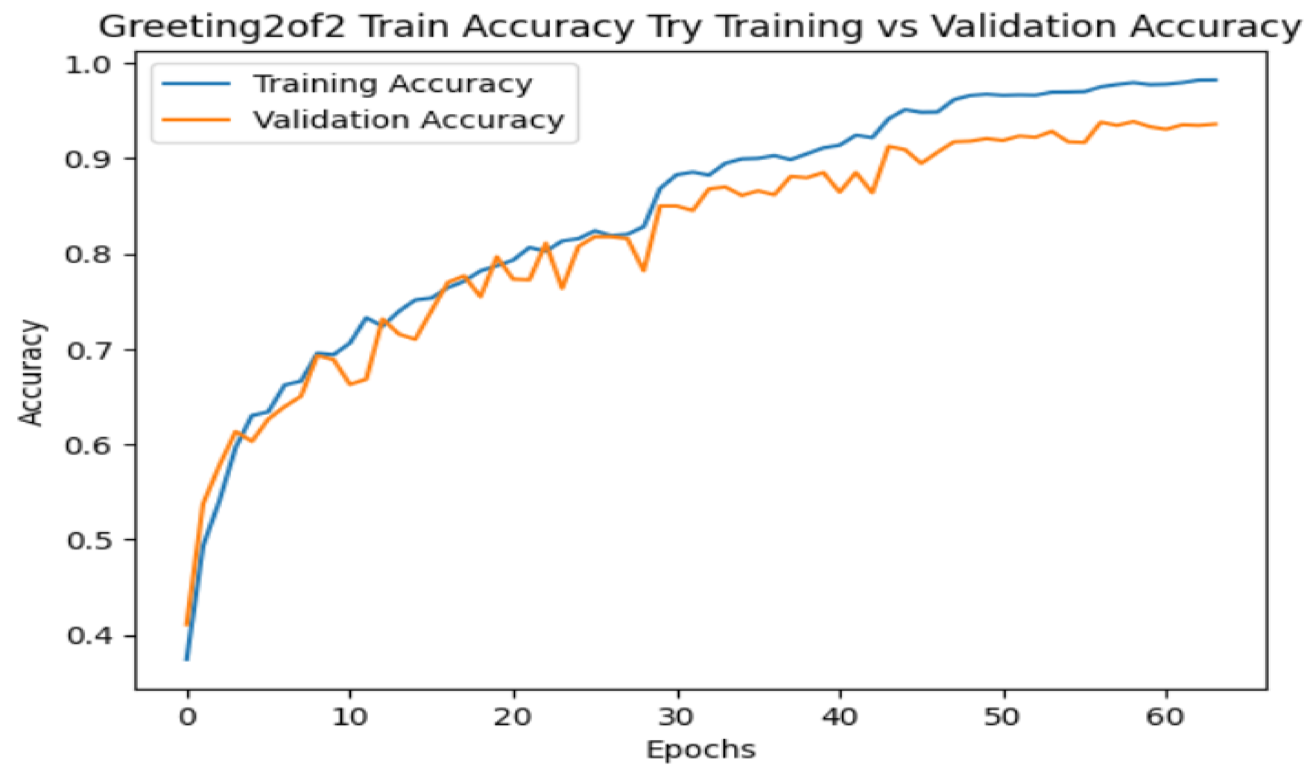

Figure 3. *Epoch vs Accuracy Graph of Best Performing Model*

Conclusion

This research successfully trained a model to predict sign language based on video inputs, achieving high validation accuracy (93.89%) along with strong performance metrics such as precision, recall, and F1 score. The study also successfully built a React Native app, which streamlined the communication between users and the Python Flask server for sign prediction. The app allows users to learn and improve their knowledge of Indian Sign Language. Future work could explore additional signs and further improve the model's accuracy by expanding the dataset and incorporating real-time datasets. Additionally, the capability to predict static images instead of video clips could be implemented, along with the ability to predict signs from a live camera feed.

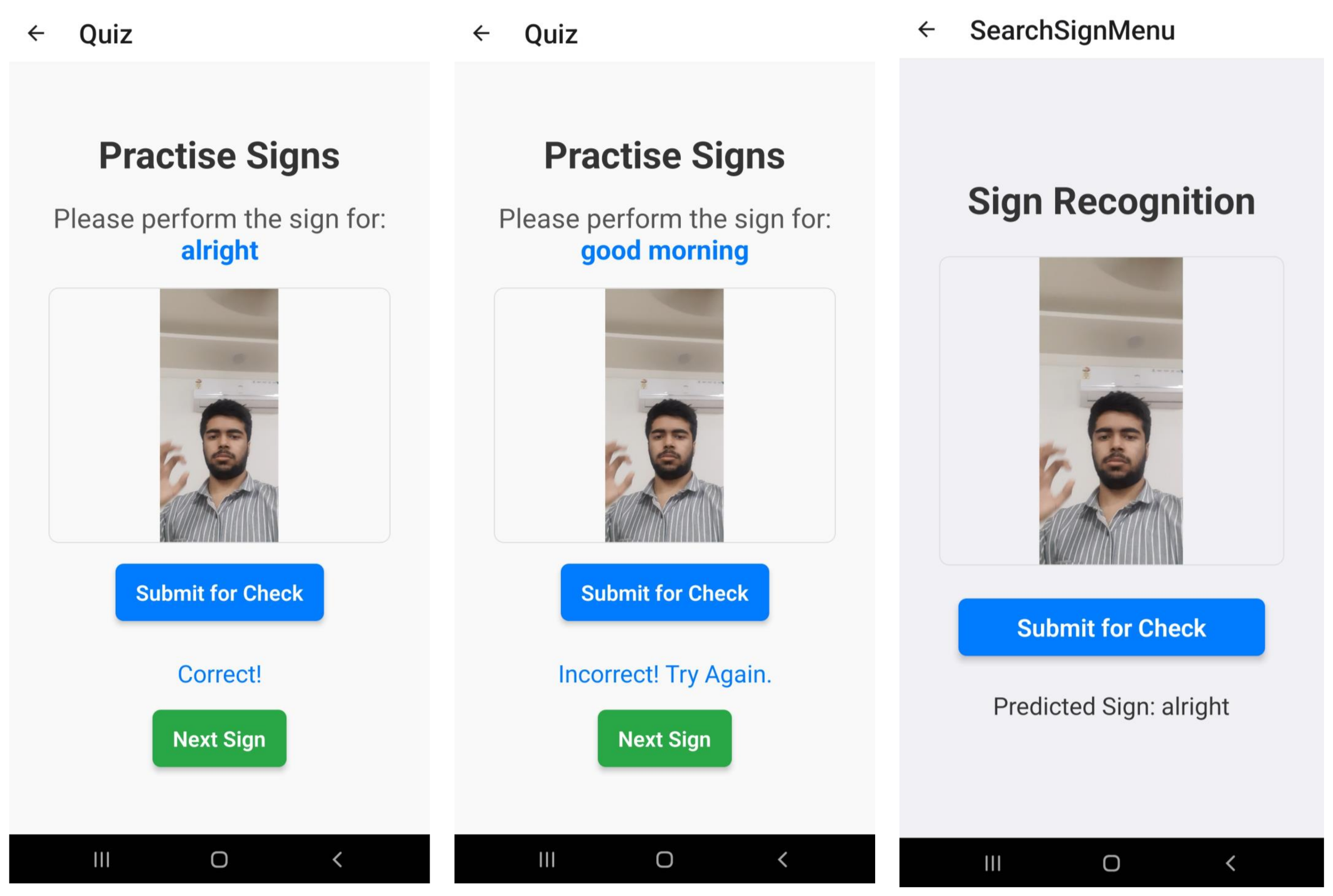

Figure 4*. App Preview*

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References

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